

Muhammad Zain-ul-Abedin

Ai / ML

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EDUCATION

Bahria University Lahore (CGPA: 3.1)
Bachelor of Science in Computer Science

Sep 2022 – Present
Lahore, Pakistan

WORK EXPERIENCE

Full Stack Developer Intern
CINQDEV

Jul 2025 – Aug 2025
Lahore, Pakistan

- Contributed to full-stack development tasks, gaining hands-on experience building web applications with modern tech stacks.
- Developed and optimized both frontend interfaces and backend logic, including APIs, databases, and responsive UIs.
- Debugged system-level bugs, enhanced application performance metrics, and successfully executed standalone tasks.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, SQL

System & Design: Node.js, FastAPI, WebSocket, REST API

DevOps & Infrastructure: Linux, Bash Scripting, SSH, Docker, GitHub, Git, GitLab

Developer Tools: Vim, grep, awk, Linux Terminal, Data Wrangling

Database & Cloud: MongoDB, PostgreSQL, SQLite, Vercel, Firebase

Frontend: React

PROJECTS

Final Year Project: AI-Based HAR With Sensors
Python, FastAPI, Deep Learning

- Developed a real-time HAR system utilizing smartphone sensor data (accelerometer, gyroscope, gravity, and orientation).
- Designed and trained a Bi-LSTM with Attention deep learning model, achieving ~95% validation accuracy across 6 activity classes.
- Built a scalable time-series data processing pipeline and integrated real-time prediction capabilities via FastAPI APIs.
- Designed a responsive web dashboard supporting live streaming data inputs, CSV file uploads, and manual system testing.

AI-Based Crypto Trading Signal Predictor
Python, Machine Learning

- Built a machine learning model using the Random Forest algorithm to evaluate historical data and predict Bitcoin buy/sell signals.
- Optimized hyperparameters and data feature sets to achieve a consistent prediction accuracy of ~60–70%.

Lexical Analyzer
C++

- Developed a custom-built compiler tool tailored for HTML, CSS, and JavaScript formatting.
- Implemented complete lexical and syntax analysis modules powered by Context-Free Grammar (CFG) and Deterministic Finite Automata (DFA) concepts.

NADRA System Using Hashing and Trees
C++

- Engineered a data-secure record system utilizing advanced data structures to mimic national identity registry workflows.

- Implemented optimized hashing and tree architectures to guarantee high-efficiency search, entry insertion, and collision handling.

Pharmacy Management System

HTML, CSS, MySQL

- Created a web-based application focused on digitalizing local hospital operational ecosystems.
- Built secure, structural schemas to seamlessly manage persistent patient records, inventory data, and transactional logs.